

CT, MRI, MRCP and HIDA in the Child's Liver.

In the last chapter we looked at the liver with ultrasound. Ultrasound is the first picture we take of almost every child with a liver problem, and very often it is the only one we need. But sooner or later every hepatology fellow is asked about the tests that come after ultrasound. Should this jaundiced baby have a HIDA scan? Would an MRCP tell us whether the bile ducts are there? Can an MRI measure the fat in this teenager's liver? Does this child with a liver mass need a CT?

These tests are very different from each other. A HIDA scan follows a radioactive tracer from the blood into the bile. An MRCP draws a picture of the fluid inside the bile ducts. Other kinds of MRI can measure how much fat or iron is in the liver, or show how a lump behaves when a contrast agent is given. And CT gives fast, sharp pictures, but at the cost of radiation.

The main idea of this chapter is simple. Each of these tests answers one question well. Most of the harm comes from asking a test a question that it cannot answer. The classic example is the cholestatic infant. Both the HIDA scan and the MRCP are very good at telling us that a baby does not have biliary atresia. Neither of them is good at telling us that a baby does have it. And yet, in many hospitals, both tests are still used to decide whether a baby goes to theatre.

So this chapter asks six questions. First, which test answers which question? Second, what can a HIDA scan really tell us in a jaundiced infant? Third, why is MRCP so difficult in a small baby, and where does it work well? Fourth, what can MRI measure, and what is special about the liver contrast agent gadoxetate? Fifth, how do we get a still infant into a scanner, and what does CT radiation cost a child? And sixth, how do the North American and European documents compare?

A short note on the sources. For the cholestatic infant, the main document is again the joint guideline of the North American and European paediatric societies, NASPGHAN and ESPGHAN, written by Fawaz and colleagues in 2017. The European Society of Paediatric Radiology, ESPR, published imaging recommendations for the jaundiced infant in 2021, written by Napolitano and colleagues. For how to perform a HIDA scan, we use the procedure guideline of the Society of Nuclear Medicine from 2010, written by Tulchinsky and colleagues. For fatty liver, we use the American liver society, AASLD, in 2025, and ESPGHAN in 2026. For the contrast agent gadoxetate, we use the American and British drug labels, both revised in 2026. And for radiation, we use the European directive known as Euratom, from 2013. As in the last chapter, most of the

numbers come not from guidelines but from single studies, and we will name them as we go.

A note on words. Sensitivity is the proportion of children with the disease whom the test picks up. Specificity is the proportion of children without the disease whom the test correctly calls normal. A test with high sensitivity is good for ruling a disease out: if it is negative, the disease is unlikely. A test with high specificity is good for ruling a disease in. Almost every test in this chapter has high sensitivity and poor specificity for biliary atresia, and that single fact explains most of what follows.

Part one. Which test answers which question.

Let us start not with the machines but with the questions. There are eight common questions in paediatric hepatology imaging, and for each one there is a best first test, a test that adds something, and a test to avoid.

The first question is whether the infant's biliary tree is open. The best path is a fasting ultrasound and then a liver biopsy. A HIDA scan helps only if it shows the tracer reaching the bowel. A direct cholangiogram, where contrast is put into the bile ducts during surgery or through the skin, settles the question. What should be avoided is using MRCP as the test that decides.

The second question is whether there is a choledochal malformation, what used to be called a choledochal cyst. Ultrasound finds it. MRCP then maps the anatomy, and in particular the common channel, the place where the bile duct and the pancreatic duct join. CT adds nothing and should be avoided.

The third question is sclerosing cholangitis. MRCP is the first test. ERCP, where a scope is passed into the duodenum and contrast is injected into the ducts, is kept for when a treatment is needed, or when the MRCP is equivocal. A diagnostic ERCP as the first test should be avoided.

The fourth question is how much fat is in the liver. The answer is an MRI measurement called the proton density fat fraction, or PDFF. Ultrasound gives a first look. CT should not be used to measure fat.

The fifth question is how much iron is in the liver. The answer is an MRI measurement of R₂ or R₂*, two numbers that rise as iron accumulates. A ferritin level on its own is not enough, because ferritin rises with inflammation.

The sixth question is what a focal lesion is. The best test is MRI with contrast. A special liver contrast agent, gadoxetate, helps to separate focal nodular hyperplasia from an adenoma, and contrast-enhanced ultrasound is an alternative. CT should be avoided unless a tumour needs staging.

The seventh question is whether there is a bile leak after surgery or transplant. A HIDA scan or an MRI with gadoxetate can both answer this, and the MRI can show exactly where the leak is.

The eighth question is the vascular map before a liver transplant. Doppler ultrasound comes first. CT or MR angiography is added when needed. What should be avoided is repeated CT.

Keep this list in mind as we go through the chapter. Each part explains one or two of these answers.

Part two. The HIDA scan.

HIDA stands for hepatobiliary iminodiacetic acid. It is the old name for a family of tracers. The tracer is injected into a vein, taken up by liver cells, and passed into the bile, just as bilirubin is. A camera then follows it. If the tracer reaches the bowel, the bile ducts between the liver and the bowel must be open.

There are two tracers in use, both labelled with technetium ninety-nine m. Mebrofenin is taken up by the liver very efficiently, about ninety-eight percent on the first pass. Disofenin is taken up a little less, about eighty-eight percent. The difference matters in a cholestatic liver, which takes up less tracer, so mebrofenin is the better choice.

The dose is based on weight, but with a minimum. If too little tracer is given, the small amount reaching the bowel may not be seen, and the scan will be falsely called non-excreting.

Many centres give phenobarbital before the scan. The Society of Nuclear Medicine guideline describes a dose of five milligrams per kilogram per day for three to five days, with ursodeoxycholic acid as an alternative. The idea is that phenobarbital increases bile flow and so makes it easier to see excretion. The cost is time: up to five days of waiting in a baby who might need a Kasai operation.

The pictures are taken continuously for the first sixty minutes, and then again at four to six hours and at twenty-four hours. Some centres add a three-dimensional view, called SPECT. Late activity in the bowel counts as excretion, which is why the twenty-four hour picture is so important.

The reading is simple. Activity in the bowel means the ducts are open. No activity in the bowel by twenty-four hours means the scan is non-excreting. The most common mistake is to read tracer in the urine, or in the stomach, as tracer in the bowel. Side views and an empty bladder help to avoid this.

And here is the most important sentence about the HIDA scan. Only a positive result, meaning excretion into the bowel, changes what we do. A non-excreting scan does not.

Why does a non-excreting scan not change what we do? Because many babies without biliary atresia also fail to excrete.

There have been several meta-analyses of the HIDA scan in biliary atresia. In every one of them, the pooled sensitivity was between ninety-three and ninety-nine percent. In other words, almost every baby with atresia has a non-excreting scan. That is why a scan that does excrete argues strongly against atresia.

The problem is specificity. In the network meta-analysis by Zhang and colleagues in 2024, which pooled forty studies, the specificity was seventy-nine percent. In the meta-analysis by Wang and colleagues in 2018, it was seventy-three percent. In Kianifar's meta-analysis of 2013, which pooled eighty-one studies, it was about seventy percent. In He's meta-analysis of 2016, it was about sixty-nine percent. And in a single-centre series of sixty-nine infants by Yang and colleagues in 2009, it was only about forty-six percent.

Put simply, between one in five and one in two babies who do not have atresia will also fail to excrete. A draining scan rules atresia out. A non-draining scan cannot rule it in.

Who are these other babies? The best answer comes from a study by Gilmour and colleagues from Toronto, published in 1997. They studied eighty-six infants up to four months of age, who had a HIDA scan and a percutaneous liver biopsy close together. Fifty-three scans were non-excreting, and thirteen of those babies did not have atresia. A quarter of the babies with idiopathic neonatal hepatitis, eight of thirty-two, had no activity in the bowel. And half of the babies with paucity of the intrahepatic bile ducts, without any obstruction outside the liver, also failed to excrete. On the other side, none of the forty babies with atresia showed excretion.

The Toronto team also tried to measure how well the liver took up the tracer, hoping that this would separate hepatitis from atresia. It did not.

The joint guideline of 2017 adds three more groups that can fail to excrete: babies with the PiZZ form of alpha one antitrypsin deficiency, babies of low birth weight, and babies on parenteral nutrition. These are exactly the babies in whom we are most likely to be worried about the liver. So a non-excreting scan tells us that the baby is cholestatic, which we already knew, and very little more.

If phenobarbital made the HIDA scan more specific, five days of waiting might be worth it. Does it?

There are two studies worth knowing. The first is a retrospective series from Bangkok by Charearnrad and colleagues, published in 2003, of ninety-five infants. In the forty-eight infants who received at least five days of phenobarbital at five milligrams per kilogram per day, the accuracy of the scan was seventy-three percent. In the eighteen infants who received no premedication at the same centre, the accuracy was one hundred percent. That is the opposite of what premedication is supposed to achieve. It was not a randomised study, so we should not read too much into it, but it certainly gives no support to phenobarbital.

The second study is the only randomised trial. Malik and colleagues from Srinagar, in India, published it in 2015. Infants were randomised to phenobarbital, to ursodeoxycholic acid, or to placebo before the scan. Among the babies who did not excrete at first, two of twenty excreted after phenobarbital, and one of ten excreted after placebo. The P value was one. In other words, there was no difference at all.

Kianifar's meta-analysis did find a higher specificity in studies that used phenobarbital. But that comparison was between different studies, done in different centres with different babies, so it is easily confounded.

The practical lesson is this. If a HIDA scan is used at all, the premedication must never delay the biopsy or the cholangiogram. Every day matters for the outcome of the Kasai operation, and five days of phenobarbital buy nothing that has been proven.

Part three. MRCP in the infant.

MRCP stands for magnetic resonance cholangiopancreatography. It uses heavily T two weighted images, in which still fluid, such as bile, is bright and everything else is dark. In adults it gives beautiful pictures of the bile ducts. In a small infant it is much harder, and there are three reasons why.

The first reason is size. Hernanz-Schulman and colleagues measured the normal common bile duct in infants in 1995. Up to three months of age, it is under one point two millimetres wide. The typical MRCP voxel, the small cube of tissue that makes up one point in the image, is about one millimetre. When a structure is the same size as the voxel, its signal is averaged with the tissue around it, and it can disappear. This is called partial volume. So a duct that is not seen may simply be a duct that is too small to see. Not seeing the duct does not mean that it is absent.

The second reason is physiology. MRCP sees fluid, and a duct without fluid in it is invisible. In an infant, there is no pressure filling the duct. A collapsed duct looks just like an atretic one. In severe neonatal hepatitis, the liver makes very little

bile, so there is little fluid to see. And a baby who has just been fed has a contracted gallbladder.

The third reason is logistics. The scan is done during free breathing, with images triggered by the breathing cycle, using a three-dimensional T two sequence. The baby must lie still for five to fifteen minutes, either after a feed and wrapped up, or under sedation. And the whole process of booking the scan, preparing the baby and getting the report may take longer than simply doing a biopsy.

The rule that follows from this is simple. If MRCP shows the whole bile tree outside the liver, including the gallbladder, the common bile duct and the connection to the duodenum, biliary atresia is excluded. If it fails to show it, that proves nothing.

How does this look in the studies? The sensitivity of MRCP for atresia was high in almost every series, between eighty-one and one hundred percent. But the specificity fell as the studies grew larger. In the first series, by Han and colleagues from Seoul in 2002, the specificity was ninety-six percent in forty-seven infants. In a study by Chen and colleagues in 2023, it was eighty-six percent in term infants and seventy-seven percent in preterm infants. In Norton's series of twenty-three infants from New York, also in 2002, it was seventy-seven percent. The pooled specificity was about sixty-five percent in He's meta-analysis of 2016 and fifty-eight percent in Wang's meta-analysis of 2018. And in the largest single series, by Liu and colleagues in 2014, which used three-dimensional MRCP in one hundred and ninety infants, the specificity was only thirty-six percent.

In Liu's series, the MRCP called atresia in one hundred and fifty-eight babies, and fifty-five of them did not have it. So the positive predictive value was sixty-five percent. One in three babies labelled as atresia by MRCP would have gone to theatre for nothing.

This pattern, excellent results in a small early series and much worse results in larger later ones, is common in diagnostic research. It is a good reason to be cautious about any single enthusiastic report.

The most useful single paper for comparing the tests is the meta-analysis by Wang and colleagues in 2018. It pooled thirty-eight studies and gave the sensitivity and specificity of six tests for biliary atresia.

Ultrasound had a sensitivity of seventy-seven percent and a specificity of ninety-three percent. The HIDA scan had a sensitivity of ninety-six percent and a specificity of seventy-three percent. MRCP had a sensitivity of ninety-six percent and a specificity of fifty-eight percent. Pale stool colour had a sensitivity of eighty-seven percent and a specificity of seventy-eight percent. Liver tests had a

sensitivity of eighty-four percent and a specificity of ninety-seven percent. And liver biopsy had a sensitivity of ninety-eight percent and a specificity of ninety-three percent.

Notice two things. First, only the liver biopsy is high on both measures. Second, the HIDA scan and the MRCP have the same shape: they catch almost every baby with atresia, but they also catch many babies without it. They are rule-out tests.

So the decision to operate should come from a biopsy or a cholangiogram, not from a HIDA scan and not from an MRCP. This is the most important practical message of the chapter.

Part four. MRCP in the older child.

Beyond infancy, the picture changes completely. The ducts are bigger, the child can often hold still, and MRCP becomes the test of choice.

For primary sclerosing cholangitis in children, Ferrara and colleagues in 2002 reported a sensitivity of eighty-one percent and a specificity of one hundred percent in twenty-one children. For a dominant stricture, that is a single tight narrowing, Rossi and colleagues in 2013 found a sensitivity of sixty-two percent and a specificity of eighty-nine percent, compared with ERCP.

A Finnish study by Vanhanen and colleagues in 2023 compared MRCP and ERCP directly, with the readers blinded. The two tests had the same accuracy for sclerosing cholangitis. ERCP showed the small peripheral ducts better, and MRCP became more accurate as children got older.

Why does it matter that MRCP is safe for repeated use? Because in adults with sclerosing cholangitis, regular imaging is used to look for cholangiocarcinoma. In children, this cancer is rare. In the large international registry reported by Deneau and colleagues in 2017, cholangiocarcinoma occurred in one percent of seven hundred and eighty-one children with sclerosing cholangitis. So routine surveillance scans in children cannot be justified by cancer risk in the same way as in adults.

For choledochal malformations, MRCP shows the shape and extent of the dilatation, and ideally the common channel between the bile duct and the pancreatic duct. But Huang and colleagues showed in 2016 that when the cyst is larger than thirty millimetres, it often hides this junction. The surgeon may then need an intraoperative cholangiogram.

The rule for the older child is: MRCP first. ERCP when a stricture needs to be brushed for cytology, dilated or stented.

Part five. What MRI can measure, and the liver contrast agent gadoxetate.

So far we have talked about pictures. But MRI can also measure. Two measurements are worth knowing.

The first is the proton density fat fraction, or PDFF. It is the proportion of the MRI signal from the liver that comes from fat, expressed as a percentage. The key paediatric study is from the NASH clinical research network, led by Middleton and colleagues in 2018, in one hundred and ten children with fatty liver disease who also had a biopsy. The mean PDFF was eleven percent in children with grade one steatosis on biopsy, eighteen percent in grade two, and twenty-six percent in grade three. PDFF separated grade one from grades two and three with an area under the curve of zero point eight seven. A cut-off of seventeen point five percent identified grade two or more with ninety percent specificity.

Just as important, the change in PDFF followed the change in histology. A fall of eleven percentage points flagged improvement, and a rise of five and a half points flagged worsening, both at ninety percent specificity. This is why PDFF is used as an endpoint in treatment trials. Both the American liver society in 2025 and ESPGHAN in 2026 treat PDFF as the non-invasive reference for measuring fat.

The second measurement is newer and comes from cholestatic infants. Lee and colleagues in 2020 studied one hundred and fifty-five infants, one hundred and ten of whom had biliary atresia. They measured the thickness of the bright tissue around the portal veins on MRI. A periportal thickness of four point two millimetres or more predicted significant fibrosis, stage two or more, with an area under the curve of zero point nine. This was better than the ratio of spleen size, with an area of zero point seven four, and better than the APRI blood score, with an area of zero point seven one.

For iron, MRI measures R two or R two star, as we said at the start. These values rise as iron builds up and allow iron to be followed without repeated biopsies.

Most MRI contrast agents stay in the blood and the fluid around the cells. Gadoxetate is different. It is taken up into liver cells by transporters called OATP, and then excreted into the bile. About twenty minutes after the injection, the normal liver becomes bright. This is called the hepatocyte phase. A lesion that contains working liver cells takes up the agent. A lesion without them stays dark. And because the agent is excreted into bile, it can show the bile ducts and any leak from them.

The dose is zero point zero two five millimoles per kilogram, which is zero point one millilitres per kilogram. That is a quarter of the dose of the usual contrast agents, so the arterial phase, the picture taken as the contrast first arrives, is weaker.

Now the important limitation. When the bilirubin is above about three milligrams per decilitre, the liver takes up much less gadoxetate, because bilirubin and gadoxetate compete for the same transporters. The images may need to be taken as late as sixty minutes. So gadoxetate cannot prove that the bile ducts are open in a jaundiced infant. It is a hepatocyte agent, not a bile test.

What about safety? Kim and colleagues in 2024 looked at one thousand six hundred and twenty-nine examinations in children. Acute adverse events occurred in one point three percent with gadoxetate and in zero point nine percent with the usual extracellular agents. The odds ratio was one point five five, and the difference was not significant. There was no safety signal.

Both the American and the British drug labels rest on the same single study of fifty-two children older than two months with liver lesions. In eighty-six percent of these children, gadoxetate added information. Yet, as we will see, the two labels come to different conclusions.

The best uses of gadoxetate in children are three. First, separating focal nodular hyperplasia, which takes up the agent, from an adenoma, which usually does not. Second, finding the exact site of a bile leak. And third, showing the margins of a hepatoblastoma before surgery. It is not the default agent, so it has to be asked for by name.

Part six. Getting a still infant, and the cost of CT.

Every MRI in a small child raises the same practical question: how do we keep the child still?

Under about three to four months of age, natural sleep usually works. The baby is fed, wrapped snugly, and put in the scanner as it falls asleep. This is called feed and wrap. Lollert and colleagues in 2024 reported a success rate of ninety-five percent in one hundred and eighty-nine neonates under four months. The image quality was three point seven on a four-point scale, against four point zero with sedation, so the difference was small. The babies who were scanned with feed and wrap were younger and lighter, and the problems with image quality increased with corrected age. After about four months, most babies need sedation or anaesthesia.

Is anaesthesia harmful to the developing brain? The best evidence is the GAS trial, reported by McCann and colleagues in 2019. Infants who had a median of fifty-four minutes of general anaesthesia had no difference in IQ at five years, compared with infants who had an awake regional anaesthetic. In 2016, the American drug agency, the FDA, issued a warning that lengthy anaesthesia, over three hours, or repeated anaesthesia in children under three years should be discussed with the family.

So a single short anaesthetic for an MRI is reassuring. But it is still not free. The practical rule is: book the MRCP only if the answer will change the plan.

CT is fast, widely available, and gives excellent pictures. It is the test of choice for many emergencies and for staging tumours. Its cost is radiation, and children are more sensitive to radiation than adults and have longer lives in which a cancer can develop.

Two large studies have changed how we think about this. The first is the British cohort of Pearce and colleagues, published in 2012, which followed one hundred and seventy-eight thousand six hundred and four children who had CT scans. Children who received a marrow dose of thirty milligray or more had a relative risk of leukaemia of three point one eight, compared with those who received less than five. Children who received a brain dose of fifty to seventy-four milligray had a relative risk of brain tumours of two point eight two.

The second is the European EPI-CT cohort, which followed six hundred and fifty-eight thousand seven hundred and fifty-two young people in nine countries, all scanned before the age of twenty-two. The excess relative risk of brain cancer was one point two seven per one hundred milligray, reported by Hauptmann and colleagues in 2023. The excess relative risk of blood cancer was one point nine six per one hundred milligray of marrow dose, reported by Bosch de Basea and colleagues, also in 2023. At the current average marrow dose of eight milligray, this means that one to two of every ten thousand children scanned will develop a blood cancer caused by the scan within twelve years.

The absolute risk is small. But it is the one risk of the test that is completely avoidable.

Not all CT scans are equal. Miglioretti and colleagues in 2013 studied paediatric CT in seven American health systems. The effective dose of a single scan ranged from zero point zero three to sixty-nine point two millisievert. Between fourteen and twenty-five percent of abdominal and pelvic scans gave twenty millisievert or more. For girls, they estimated one solid cancer for every three hundred to three hundred and ninety abdominal and pelvic scans. And if the highest quarter of doses were reduced to the median, forty-three percent of the projected cancers would be prevented.

The lesson is the spread. Use a child-sized protocol. Scan one phase, not three. And do not do a CT when ultrasound or MRI can answer the question.

Nuclear medicine is not free of radiation either. Grant and colleagues in 2015 compared the North American and European dose recommendations for children. In sixty-two percent of cases in children aged five or younger, the

European dosage card gave more tracer than the North American guidance. So the two continents do not even agree on how much activity to give.

Part seven. What the guidelines say.

How do the North American and European documents compare? There are nine points worth knowing.

First, the HIDA scan in the cholestatic infant. Here they differ. The joint NASPGHAN and ESPGHAN guideline of 2017 says that a selective HIDA scan may help to exclude atresia, with a grade of one B. The European paediatric radiologists, ESPR, in 2021, say that the HIDA scan is not useful for atresia.

Second, MRCP in the cholestatic infant. Here they are aligned. The joint guideline gives it a limited role because of its low specificity, and ESPR says it is not useful for atresia.

Third, the equivocal case. Here they differ. The joint guideline of 2017 favours the liver biopsy, and gives only a limited role to ERCP or a cholangiogram through the skin, called PTCC. The ESPR flowchart of 2021 goes to PTCC or ERCP, together with a biopsy.

Fourth, the dose of tracer for nuclear medicine in children. Here they differ. North America uses a consensus from 2016, updated in 2024. Europe uses the dosage card of the European nuclear medicine association, EANM, which, as we saw, often gives more.

Fifth, gadoxetate in children. Here the drug labels are split. The American label, revised in 2026, says that it is safe and effective in children, including term neonates. The British label, also revised in 2026, says that its safety has not been established under eighteen years. Both labels rely on the same fifty-two children. So the same agent is on label in a baby in Boston and off label in a teenager in London.

Sixth, the older linear gadolinium agents. The labels are split again. In 2017, the FDA added a warning about gadolinium retention in the body, but did not restrict these agents. In the same year, the European Medicines Agency suspended the intravenous linear agents, but kept the liver-specific ones, including gadoxetate.

Seventh, the governance of CT dose. Here Europe adds something. In North America, the Image Gently campaign is voluntary. In Europe, the Euratom directive of 2013 makes diagnostic reference levels binding and requires the dose to be recorded in the report.

Eighth, measuring fat in the liver. Aligned. The American liver society in 2025 calls MRI PDFF the only validated method, and ESPGHAN in 2026 calls it the non-invasive reference.

Ninth, suspected sclerosing cholangitis. Aligned. Both the American liver society in 2022 and the European liver society in 2022 put MRCP first.

So the documents agree that MRCP does not decide atresia, that PDFF measures fat, and that MRCP comes first for sclerosing cholangitis. They disagree on the HIDA scan, on the equivocal infant, on how much tracer to give, and on the labels of gadolinium agents.

Part eight. Eight things to remember, and from question to test.

Let us bring this together into eight pearls.

First, a HIDA scan that excretes argues strongly against atresia. A scan that does not excrete is non-specific, with a specificity between forty-six and seventy-nine percent.

Second, babies with paucity of the bile ducts, neonatal hepatitis, PiZZ alpha one antitrypsin deficiency, prematurity or parenteral nutrition can all fail to excrete.

Third, phenobarbital premedication costs five days and has no proven benefit in a randomised trial.

Fourth, the infant bile duct is about one millimetre wide and has no filling pressure. On MRCP, seeing it excludes atresia. Missing it proves nothing.

Fifth, beyond infancy, MRCP replaces diagnostic ERCP for sclerosing cholangitis and for the anatomy of choledochal malformations.

Sixth, MRI PDFF grades fat and tracks change. In cholestatic infants, a periportal thickness of four point two millimetres or more suggests significant fibrosis.

Seventh, gadoxetate does not work well when the bilirubin is above three milligrams per decilitre, and the American and British labels disagree about its use in children.

Eighth, abdominal CT gives the highest dose of any paediatric scan. Justify it, size it for the child, and scan one phase.

To finish the teaching part of the chapter, here are the common clinical scenarios, and for each one, the first test, the next step if the answer is unclear, and what not to do.

A cholestatic infant with pale stools. Start with a fasting ultrasound, a GGT and a liver biopsy. If the answer is still unclear, do a cholangiogram, during surgery, through the skin, or by ERCP. Do not wait five days for phenobarbital.

A cholestatic infant who has already had a HIDA scan. If it excretes, atresia is unlikely. If it does not, do the biopsy anyway. Do not call atresia on a non-excreting scan.

A cyst at the porta hepatis on ultrasound. Do an MRCP. If it is still unclear, the surgeon will do a cholangiogram during the operation. Do not do a CT.

A teenager with inflammatory bowel disease and a high GGT. Do an MRCP. If it is normal, a liver biopsy looks for small duct sclerosing cholangitis. Do not start with a diagnostic ERCP.

An obese child with a raised ALT. Start with ultrasound, and add MRI PDFF if a number is needed. Biopsy if the answer is still unclear. Do not use CT.

A liver mass. Start with ultrasound, then MRI with contrast. Add gadoxetate MRI or a biopsy if needed. Avoid multiphase CT of the abdomen.

A bile leak after transplant. Ultrasound looks for a collection. A HIDA scan or a gadoxetate MRI shows the leak.

An infant who needs an MRI. Under four months, try feed and wrap. Otherwise, a short sedation. And do not book it if the result will change nothing.

Four clinical cases.

Case one. A five-week-old boy has jaundice and pale stools. His conjugated bilirubin and GGT are raised. The ultrasound after a four-hour fast shows a small gallbladder. The team orders a HIDA scan with five days of phenobarbital first. On day six, the scan shows no excretion at twenty-four hours, and the team plans a Kasai operation for the next week. What do you think? Two things are wrong. First, the phenobarbital cost five days and has no proven benefit. Second, a non-excreting scan does not prove atresia. In the Toronto series, a quarter of babies with neonatal hepatitis and half of babies with bile duct paucity also failed to excrete. This baby needs a liver biopsy now, and if the biopsy is suggestive, a cholangiogram at the start of the operation. The HIDA scan could only have helped if it had shown excretion.

Case two. A seven-week-old girl, born at thirty-two weeks and fed by parenteral nutrition for the first month, has conjugated jaundice. An MRCP under sedation does not show the common bile duct. The report says "findings suggestive of biliary atresia". How much weight should this carry? Very little. Her common bile duct is probably under one point two millimetres, which is about the size of one

MRI voxel. There is no filling pressure, and a cholestatic liver makes little bile. In Liu's large series, the specificity of MRCP was only thirty-six percent, and one in three babies labelled atresia did not have it. Preterm birth and parenteral nutrition make a false positive more likely. If the MRCP had shown the whole duct system, atresia would have been excluded. Since it did not, the next step is still a biopsy, and a cholangiogram if needed.

Case three. A fourteen-year-old boy with ulcerative colitis has a GGT three times the upper limit. The gastroenterology team asks for an ERCP to look for sclerosing cholangitis. What would you suggest instead? An MRCP. In the blinded Finnish comparison, MRCP and ERCP were equally accurate for sclerosing cholangitis, and MRCP improves with age, so at fourteen it should perform well. Both the American and European liver societies put MRCP first. ERCP should be kept for treatment, for example if a dominant stricture needs to be brushed, dilated or stented. If the MRCP is normal and the GGT stays high, a liver biopsy can look for small duct disease, which MRCP cannot see.

Case four. A nine-year-old girl has a four-centimetre liver lesion found on an ultrasound done for abdominal pain. Her liver tests are normal. The emergency team suggests a three-phase CT. What would you do? The best test is MRI with contrast, and here it is worth asking for gadoxetate by name. If the lesion takes up gadoxetate in the hepatocyte phase, it is most likely focal nodular hyperplasia, and it may not need a biopsy. If it does not, an adenoma or another lesion is more likely. A three-phase abdominal CT would give her a large dose of radiation. In the American data, abdominal and pelvic CT in girls led to about one solid cancer for every three hundred to three hundred and ninety scans. Mention to the family that gadoxetate is licensed for children in the United States, but not in the United Kingdom, so in some countries its use would be off label.

Key points to remember.

Start from the question, not the machine. Each test answers one question well.

A HIDA scan that excretes argues strongly against biliary atresia. A scan that does not excrete is non-specific, and between one in five and one in two babies without atresia also fail to excrete.

Neonatal hepatitis, bile duct paucity, PiZZ alpha one antitrypsin deficiency, prematurity and parenteral nutrition can all give a non-excreting scan.

Phenobarbital before a HIDA scan costs five days, and the only randomised trial found no benefit. It must never delay the biopsy.

The infant bile duct is under one point two millimetres wide. On MRCP, seeing the whole duct excludes atresia. Not seeing it proves nothing.

The specificity of MRCP for atresia fell from ninety-six percent in the first small series to thirty-six percent in the largest one.

In the meta-analysis of six tests, only liver biopsy was high on both sensitivity and specificity. The decision to operate comes from a biopsy or a cholangiogram.

Beyond infancy, MRCP is the first test for sclerosing cholangitis and choledochal malformations. ERCP is for treatment.

MRI PDFF measures fat and tracks change. A periportal thickness of four point two millimetres or more suggests significant fibrosis in cholestatic infants.

Gadoxetate is a hepatocyte agent, not a bile test. It fails when the bilirubin is above three milligrams per decilitre. Use it for focal nodular hyperplasia versus adenoma, bile leaks and hepatoblastoma margins.

Under four months, try feed and wrap. A single short anaesthetic did not affect IQ in the GAS trial, but book the MRI only if it will change the plan.

CT in childhood slightly increases the risk of leukaemia and brain tumours. Abdominal CT is the highest-dose scan. Justify it, size it and scan one phase.

North America still allows a selective HIDA scan, and European radiologists do not. The American and British labels disagree about gadoxetate in children.

Questions to test yourself.

Question one. A HIDA scan in a cholestatic infant shows tracer in the bowel at six hours. What does this tell you? Answer: it argues strongly against biliary atresia, because the sensitivity of the scan for atresia is between ninety-three and ninety-nine percent.

Question two. Name five conditions, other than atresia, that can give a non-excreting HIDA scan. Answer: idiopathic neonatal hepatitis, paucity of the intrahepatic bile ducts, PiZZ alpha one antitrypsin deficiency, low birth weight, and parenteral nutrition.

Question three. What did the only randomised trial of phenobarbital before a HIDA scan show? Answer: no benefit. Two of twenty non-excreting babies excreted after phenobarbital and one of ten after placebo, with a P value of one.

Question four. Give three reasons why MRCP may fail to show the bile duct in a young infant who does not have atresia. Answer: the duct is under one point two millimetres, about the size of a voxel, so partial volume can erase it. There is no filling pressure, so the duct may be collapsed. And in severe hepatitis the liver makes little bile.

Question five. In the meta-analysis by Wang and colleagues, which test for atresia was high on both sensitivity and specificity? Answer: the liver biopsy, with a sensitivity of ninety-eight percent and a specificity of ninety-three percent.

Question six. Why can gadoxetate not be used to prove that the bile ducts are open in a jaundiced infant? Answer: because when the bilirubin is above about three milligrams per decilitre, the liver takes up much less gadoxetate, so it may not be excreted into the bile even if the ducts are open.

Question seven. What PDFF value identified grade two steatosis or more with ninety percent specificity in the paediatric study by Middleton and colleagues? Answer: seventeen point five percent.

Question eight. What does the Euratom directive require for CT that the American Image Gently campaign does not? Answer: binding diagnostic reference levels, and the recording of the dose in the report.

End of chapter.